

Incidence of Cataract and Outcomes after Cataract Surgery in the First 5 Years after Iodine 125 Brachytherapy in the Collaborative Ocular Melanoma Study

COMS Report No. 27

Collaborative Ocular Melanoma Study Group*

Purpose: To evaluate the effect of the radiation dose to the lens on cataract formation and effect of cataract surgery on visual acuity (VA) among patients with choroidal melanoma treated with iodine 125 (I^{125}) brachytherapy.

Design: Prospective study of patients enrolled in one arm of a randomized clinical trial.

Participants: Patients enrolled in the Collaborative Ocular Melanoma Study (COMS) who received I^{125} brachytherapy as randomly assigned and also were phakic, with no history of cataract in the study eye at the time of enrollment (n = 532).

Methods: Each follow-up examination included an interim ocular history and full ophthalmic examination. Lens status (phakic, pseudophakic, or aphakic) and best-corrected VA were recorded. For the purpose of this analysis, the date of the first examination at which an eye was reported to be aphakic or pseudophakic, to have vision-limiting lenticular opacities, or to have had cataract surgery was defined as the observed time of cataract development. Date of cataract surgery was defined as the date of the first follow-up examination at which cataract surgery was reported.

Main Outcome Measures: Incidence of cataract and outcomes after cataract surgery.

Results: During the first 5 years of follow-up, cataracts developed in 362 (68%) of the 532 study eyes, including 49 (9%) that had had cataract surgery. By 5 years, 83% of study eyes were reported to have a cataract (95% confidence interval [CI], 79%–87%), and 12% had undergone cataract surgery (CI, 9%–15%) in the study eye. Eighteen percent of eyes that received a dose of 24 Gy or higher to the lens underwent cataract surgery, whereas only 4% of patients with <12 Gy to the lens underwent cataract surgery. Median VAs were 20/125 before cataract surgery and 20/50 after cataract surgery. After cataract surgery, VA improved by 2 lines or more in 32 (66%) patients and remained stable in 13 (26%) patients. The most common cause of lack of visual improvement after cataract surgery was presence of radiation retinopathy.

Conclusion: Although cataract surgery was infrequent among COMS patients, VA remained stable or improved in the majority of these eyes after cataract surgery. *Ophthalmology* 2007;114:1363–1371 © 2007 by the American Academy of Ophthalmology.



Enucleation and brachytherapy commonly are used to manage uveal melanoma.^{1–5} Since the first use of brachytherapy, many different radioisotopes, including gold 198, iodine 125 (I^{125}), ruthenium 106/rhodium 106, iridium 192, and

palladium 103 (Pd^{103}), have been used. With increased use of brachytherapy, assessment of local side effects and their influence on visual acuity (VA) has become particularly important.

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*See "Appendix" for Study Group information.